

PRODIGY_GA_02 - Text-to-Image Generation using Stable Diffusion

Task Description

Build a text-to-image generation application using a pre-trained generative AI model such as Stable Diffusion. The model generates realistic images based on textual prompts provided by the user.

Objective

To understand and implement generative AI models capable of creating visual content from natural language descriptions.

Technologies Used

- Python
- Stable Diffusion v1.5
- Diffusers Library
- PyTorch
- Hugging Face
- Google Colab

Features

- Generate images from text prompts
- Utilizes a pre-trained Stable Diffusion model
- Supports multiple creative prompts
- Saves generated images automatically

Sample Prompts

1. An AI engineer working in a futuristic lab
2. A cyberpunk city glowing at night
3. A humanoid robot studying in a digital library

How It Works

1. The user provides a text prompt.
2. The Stable Diffusion model interprets the prompt.
3. The model generates an image based on the textual description.
4. The generated image is displayed and saved.

Output

The model successfully generates high-quality images corresponding to user-provided prompts.

Learning Outcomes

- Understanding Generative AI
- Working with pre-trained diffusion models
- Image synthesis from natural language prompts
- Using Hugging Face Diffusers library
- GPU-accelerated AI inference using Google Colab

Internship

This project was completed as part of the Generative AI Internship at Prodigy InfoTech.